

Computer Architectures notes

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1 Lesson 22-09-26

1.1 Introduction

The aim of this course is to provide basic knowledge about the architecture of modern *microprocessor*-based systems. Special emphasis will be put on the *microprocessor core* itself.

The course is structured in two main blocks:

1. Modern processor architecture. This will cover computer design, the *RISC* processor architecture, superscalar processors and disrupting architectures;
2. *ARM*-based systems, that is *ARM* assembly language and *ARM Cortex-M3* based systems. Both development boards and emulation systems will be used to aid in the study of *ARM*-based systems.

1.2 CISC vs. RISC

We first go over what an **ISA** (*Instruction Set Architecture*) is in the context of processor design. Then we distinguish between **CISC** (*Complex Instruction Set Computer*) and **RISC** (*Reduced Instruction Set Computer*) architectures.

1.2.1 High level vs. Low level languages

Many programming languages exist. They can be classified into:

- **High-level** programming languages, which abstract over the machine to offer more sophisticated capabilities. This gives developers the freedom to express the same algorithm in many ways, and usually to express the same algorithm with significantly less lines of code than those used in low-level languages;
- **Low-level** programming languages, which stay closer to an **ISA** (*Instruction Set Architecture*). An ISA is a specification which defines:
 - Supported instructions;
 - Data types;
 - Registers;
 - Addressing modes.

In low-level languages, operations are smaller than those provided in high-level languages, so even simple algorithms usually take more instructions to express.

We can express this difference by taking the simple example of a loop:

1. In high-level languages such as *Python* and *C*, loops are expressed via a simple and predefined syntax which allows to compactly express iteration value initialization, update and the exit condition;
2. In a low-level language, all of these operations are distinct: the programmer has to explicitly initialize an iteration value, perform a comparison to check for the exit condition, and manually update the iteration value.

Another idea to note is that high-level languages are usually designed to be *portable*, that is *interpreted* or *compiled* for a variety of systems. On the other hand, low-level

languages are usually designed for a single machine (a single type of processor), and porting to other platforms requires manual conversion work.

Programs written in high-level languages are usually compiled on the machines they are meant to run on: this means they are converted to low-level machine code which can be directly executed. For environments where running a full compiler is not feasible (such as a development board), a technique known as *cross-compilation* can be used: the program is compiled on a second machine, into machine code for the first machine. All of this of course doesn't apply to low-level work, where we are expected to write the same routines from scratch, and likely differently, for each target architecture.

1.2.2 Data types and addressing

An ISA also defines the **data types** the CPU can address. For instance, the RISC architecture might allow a programmer to address bytes, half words (double bytes) and words (double half words). With this being said, we note that usually low-level languages interpret all data as *binary* data (complex types such as *integer*, *float*, ... are usually added in as high-level constructs).

There are two main methods of performing addressing:

- **Register-Memory** architectures: operands of instructions can be immediates (contained within the instruction encoding), or contained in memory *or* in registers. If *all* operands can be in registers or in memory (or in combinations), the architecture is called *register-plus-memory*. Clearly, as memory accesses are also operations that have to be carried out by the processor, these kinds of ISAs are more complex.
- **Register-Register** architectures: these are also called **load-store** architectures. Instructions of register-register architectures are explicitly divided into:
 - Memory access (*load* and *store* between memory and registers);
 - **ALU** (*Arithmetic and Logic Unit*) operations (only between registers). The ALU itself is one of the most important components of a processor.

It follows that operands of ALU instructions are immediates or contained in registers.

In CISC (usually *register-to-memory*) instruction sets, the order of instructions can affect the number of memory accesses. The more the number of memory accesses increases, the more performance degrades. Load-store architectures, by their nature, do not suffer from this problem

1.2.3 CISC

CISC (*Complex Instruction Set Computer*) was the name given to the first computer designs. The main purpose is to complete a task in as few assembly lines as possible. This makes sense in an environment where programmers themselves are tasked with writing assembly code, as we said earlier that low-level languages tend to express the same concepts with more lines of code (we refer to this as a *semantic gap* between low and high-level languages).

Common examples of CISC CPUs are:

- x86 (1978) and x64 (2003) family, by Intel and AMD;
- Motorola 68000 series (1979);
- IBM System/360 family (1964);
- PDP-11 (1970) and VAX families (1977), by DEC.

Some characteristics of CISC are:

- Instruction encoding has *variable length*: for an immediate instruction we need to store the immediate, whereas for register to register or register to memory instructions we need to store register and memory addresses. As an example, x64 family instructions are between 1 and 6 bytes;
- The processor architecture has to be more *complex*, as a single instruction can express complex operations;
- The number of *clock cycles* required for execution of a single instruction is wildly variable: if an instruction requires a memory access, we need to wait for the memory to respond, which may happen on timescales far larger than those of the CPU;
- The resulting *code size* is far smaller than that of other architectures.

1.2.4 RISC

RISC (*Reduced Instruction Set Computer*) is the name given to a family of processors initially developed during the 1980s. The main purpose is to make the processor as simple as possible: this requires more work on the programmers' side, but allows the CPU to move much faster and execute many more instructions in the same amount of time as a CISC CPU. The limitation of more complex code, then, can be removed by making use of highly optimized *compiler* which take the toll of manual assembly programming away from programmers.

Common examples of RISC CPUs are:

- ARM family (1985), designed by ARM Holdings, which will also be the focus of the second part of this course;
- MIPS family (1984), by MIPS Technologies Inc.;
- RISC-V (2014), an example of a *register-to-register* ISA;
- SPARC family (1985), by Sun Microsystems;
- PowerPC (1991), by Apple, IBM, and Motorola.

Some characteristics of RISC are:

- The instructions themselves are *simpler*;
- The length of the instructions can be *fixed* (for example, in RISC-V they have a fixed size of 32 bits). A fixed instruction size sometimes requires some bits of the instruction encoding to remain unused: this is still preferred over variable length encoding as it allows the component of the CPU that performs the *fetch-decode* cycle to be far simpler;
- The processor architecture is *simpler*;
- The number of clock cycles required to execute an instruction is normally *lower* than in CISC, and there is less overall variance. This makes it easier to design and validate both RISC code and processors, as the amount of time taken by each instruction is easy to estimate (optimistically, 1 cycle per instruction).

Common factors that disallow having exactly 1 cycle per instruction involve:

- *Latencies* introduced by CPU optimizations, such as pipeline flushes and branch misses;
- *Memory accesses*: load and store instructions always have to at least wait for the cache, and if a miss happens, for the memory, which we said is usually slower than the CPU;

- Some *complex instructions* can be introduced back into the architecture: these give the programmer more power, but also require more clock cycles to complete.
- The resulting *code size* is larger than that of other architectures.

All of these characteristics contributed to making RISC architectures more popular than CISC architectures nowadays. In fact, some modern CISC processors actually convert the CISC instructions to an internal RISC representation, which then gets actually executed on the processor.

Also, the tradeoff given by longer code length is balanced by the fact that we are allowed to have *easier* CPUs, that is CPUs that are easier to *design* and *validate*. Some extensions have been introduced (such as the C "Compressed" extension for RISC-V) to make RISC-V instructions smaller and reduce the code memory footprint.

1.2.5 Comparison of addressing modes

As an example, let's compare the available addressing modes in x86 (a CISC architecture) and RISC-V (a RISC architecture). We then go over a specific detail we will encounter in ARM architectures.

- In **x86**, a number of addressing modes are available:
 - Register mode: `MOV AX, BX`
 - Immediate mode: `MOV AX, 2000`
 - Displacement or direct mode: `MOV AX, [DISP]`
 - Register indirect mode: `MOV AX, [DI]`
 - Based indexed mode: `MOV AX, [BX + DI]`
 - Indexed mode: `MOV AX, [SI + 2000]`
 - Based mode: `MOV AX, [BP + 0100]`
 - Based indexed displacement mode: `MOV AX, [SI + BP + 2000]`

Such a large amount of modes could look a bit confusing. The idea is that modern x86 processors support **SIB** (*Scale-Index-Base*) addressing for arrays, *Mod R/M* bits, which allow for the following parameters:

- **Base** is a general-purpose register, containing a base memory address;
- **Index** is a general-purpose register containing a memory offset;
- **Scale** is a special constant, which can be 1, 2, 4, or 8;
- **Displacement** is an arbitrary constant, offered as an immediate in the instruction.

From these parameters, the processor computes an *Expected Address (EA)* is generally defined as:

$$EA = \text{Base} + \text{Index} \times \text{Scale} + \text{Displacement}$$

Any of these components can be omitted, resulting in the complex array of addressing modes shown above (the indexed/based move differ mainly in semantics because of historical reasons).

In the ISA, these addressing modes are specified by the instruction encoding, by the *ModR/M* byte, with an optional *SIB* byte. For more information, refer to the notes available at this very long URL: <https://raw.githubusercontent.com/seggiani-luca/appunti-rl/34228f66db395637bd1824d04f3130b977cc0ce4/master/master.pdf> (in italian).

We show one last table to summarize:

Addressing mode	Example	Loaded address
Register	ADD AX, BX	AX
Immediate	ADD AX, FFh	FFh
Displacement	ADD AX, [FFh]	mem[FFh]
Based	ADD AX, [BX]	mem[BX]
Indexed	ADD AX, [CX]	mem[CX]
Scale indexed	ADD AX, [CX*4]	mem[CX × 4]
Indexed based	ADD AX, [BX + CX]	mem[BX + CX]
Scale indexed based	ADD AX, [BX + CX*4]	mem[BX + CX × 4]

- In **RISC-V**, the same operations is significantly simpler. We have available:
 - ALU operations:
 - * Register addressing mode: **ADD** t0, t1, t2
 - * Immediate addressing mode: **ADDI** t0, t1, 10

with the first register taken as the destination, in common RISC fashion.
 - Load / store operations, in base + displacement addressing mode: **LW** t2, 4(t1), where the constant before the open parenthesis is an immediate offset, and the register contains a memory address. Of course, since for instant RV32x is a 32 bit architecture, 32 bit instructions cannot fully contain immediate addresses: this is taken into account by special facilities in the processor which allow full address loading via multiple instructions.
- The same is available in ARM, with the distinction of having *pre-indexed* and *post-indexed* memory operations, that is control over the time when the base register is updated.

Being more specific, ARM allows load and store instructions to optionally *alter* the address contained within a base register. 3 forms are available:

 - **ldr** r0, [r1, #4] is equivalent to immediate offset addressing in RISC-V, with #4 being the immediate operand;
 - **ldr** r0, [r1, #4]!, with the additional !, means performing a *write-back* or **pre-indexing** operation. The calculated address, obtained by adding the offset, is *written back* into the base register before the load operation. The result we get is that of a load from the base address plus the offset, with the base address reflecting the change after execution. Compared to RISC-V, this saves an instruction;
 - **ldr** r0, [r1], #4 means performing a **post-indexing** operation. This means increasing the value of the base register by offset, after performing the load. We can think of this as being analogous to the ++ syntax available in C. This also saves an instruction compared to RISC-V.

As we can see, the main point of this complication is to save a few instructions when accessing arrays, or generally where striding memory access patterns are needed in ARM architectures.

2 Lesson 23-09-26

2.1 Processor design

The first general-purpose computer was created in the late 50s. A Personal Computer, that can now be bought for about \$500, is practically equivalent (in terms of performance and memory) to what could be bought for about \$1M in 1985.

This evolution has been made possible by:

- Advances in semiconductor technology
- Innovations in computer architecture
- Advances in software.

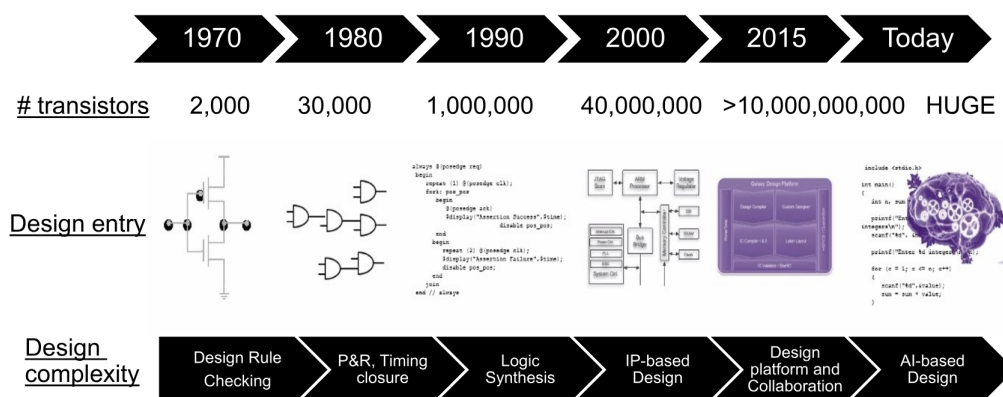
Despite this the main architecture used today is still the same that was first applied to real computers, that is the *Von Neumann Architecture*.

2.1.1 Advanced architectures

Of course, there are modern advancements which have to be noted. We refer to these as *advanced architectures*:

- **Advances in computer architecture:** performance has been improved without fundamentally changing the Von Neumann model through techniques such as pipelining, which overlaps the execution of multiple instructions, and superscalar execution, which allows multiple instructions to be executed in parallel in a single clock cycle. Other major improvements include caches, branch prediction, and out-of-order execution.
- **Advances in software and tooling:** compilers have become increasingly sophisticated at optimizing programs for the underlying hardware, exploiting instruction-level parallelism, registers, caches, and specialized instructions. Better operating systems, programming languages, debuggers, profilers, and development tools have also made increasingly complex hardware practical to use.

Chip-design history



2.1.2 Moore's law

Definition 2.1: Moore's law

The number of devices (i.e., transistors) that can be integrated into a single chip doubles every 18/24 months.

Despite the name, Moore's law describes an empirical relationship, not a scientific law. This type of observation, an experience curve effect, quantifies efficiency gains from learned experience in production. In fact, Moore's empirical evidence did not imply that the historical trend would continue; nevertheless, his prediction has held since 1975 and has since become known as a law.

We can also say that thanks to Moore's law, we are today able to integrate more and more transistors within one circuit, giving many powerful options to designers.

Many technologies are available to streamline chip design (for instance, *synthesis* and automated tools). Having more transistors within a single processor also allows for advanced features. For instance, something like a *floating point* processing unit executing alongside the main processor takes a large number of transistors to implement, which larger chip transistor densities allow.

2.1.3 Processor performance evolution

Processor **performance** can be measured by counting the number of operations executed or the time needed to execute them.

- User point of view: performance = response time (time between start and completion of an operation);
- System manager point of view: performance = *throughput* (total amount of work done in the time unit).

Processor performance evaluation is often performed by letting the computer to execute applications and observing its behavior. Unfortunately, the choice of the applications severely affects the performance. In the ideal case, one should use as workload the mix of applications the user will run. However, they are normally unknown, and largely variable from one user to another. Therefore, some **benchmarks** are selected to mimic real cases.

Processor performance has steadily increased since the 1970s, where the first main-frame machines started being ranked based on performance.

- From the 1970s to around the 1980s, yearly performance increases were of about 25% ~ 30%. We have to recall that this was a time when mainframes and mini-computers dominated the market: performance improvements were driven mainly by advances in integrated-circuit technology, together with architectural improvements such as larger register sets, more buses, caches, and pipelining.
- In the 1980s, the introduction of **RISC** (*Reduced Instruction Set Computer*) architectures, as opposite to **CISC**, led to a yearly performance gain of around 50%. These allowed for major processor redesigns, which enabled optimization unfeasible (at least at the time) on CISC architectures;
- In the first years of the 2000s, processor performance growth hit a slow down because aiming for higher and higher clock speeds became unfeasible. There were 3 main reasons:

- **Power** issues: higher clock speeds meant higher power dissipation, which cooling could only partially keep under control;
- Lower **ILP** (*Instruction Level Parallelism*): this was disadvantageous for the *superscalar* architectures used at the time, characterized by high pipeline depth and instruction level parallelism;
- Unchanged **memory latency** remained a big bottleneck.
- The solution most vendors adopted from the mid 2000s onwards, and which to this day is widely adopted, is *multi-core* processors. In fact, since 2004, major industries stopped the race for higher-performance single processor projects running at higher clock frequencies, and embraced *multi-core* CPUs.
In a **multi-core** processor, we effectively have multiple processors (or processor *cores*) running in parallel on a single, or multiple tasks. Note that multi-core setups increase *throughput*, and not *response time*.
- In more recent times (mid 2010s) we have seen a 12% year improvement, or doubling every 8 years (mainly due to the limits on instruction parallelism). This has slowed down to a 3.5% performance improvement, or doubling every 20 years, in the last 10 years (at the time of writing).

Today, we can not only talk about processor improvements, but the fact that modern computer systems employ various types of *accelerators*, that is circuits intended to support specific applications alongside the CPU, like graphics or AI (an example is given by **NPU**s, *Neural Processing Units*).

In the recent years, the widespread adoption of Artificial Intelligence (AI) applications prompted the introduction of new computing architectures denoted as AI accelerators, such as **GPU**s (*Graphics Processing Units*). GPUs were originally introduced for graphics applications, but their current success is due to their ability to effectively manage large and regular data structures. The GPU market (in 2024 corresponding to about 18 B\$), is expected to grow to 406 B\$ by 2033 (with a Compound Average Grow Rate, or CAGR, of 41,39%). The CAGR of the CPU market is about 7%.

2.1.4 Costs

In processor, and more generally computer design, **costs** have to be taken into account. These can be divided into two main categories:

- **Design costs:** these are largely *fixed* or *non-recurring* costs, usually referred to as **NRE**s (*Non-Recurring Engineering costs*). They include activities such as architecture design, circuit design, verification, physical design, software development, and mask production. NRE costs grow rapidly with chip complexity and with the sophistication of the manufacturing process. In particular, moving to a newer process node requires expensive new design tools, masks, **IP** (*Intellectual Property*), and extensive verification. Since these costs are paid before mass production, they have to be amortized over the number of chips sold. Therefore, a high-volume product can tolerate large NRE costs because they are distributed over many units, whereas for a low-volume product they can represent a significant portion of the final cost;
- **Manufacturing costs:** these are the *recurring* costs associated with producing each chip. They include wafer fabrication, packaging, assembly, and testing. The semiconductor manufacturing process itself requires extremely expensive fabrication facilities (*fabs*), whose cost increases substantially for advanced process nodes.

An important factor when evaluating manufacturing costs is the **yield**, i.e., the percentage of manufactured chips that satisfy all specifications and pass the testing phase. Not every die on a wafer is functional: manufacturing defects can make some chips unusable. Consequently, a low yield increases the effective cost of every working chip. Yield normally improves over time as the manufacturing process matures. This phenomenon is known as the **learning curve**: defects are progressively reduced, manufacturing parameters are optimized, and the production process becomes more reliable. As yield increases, the effective cost per functional chip decreases.

Testing and validation also contribute significantly to the recurring cost of a processor. Complex processors require extensive electrical testing and, in many cases, additional screening and characterization before they can be sold. Thus, manufacturing cost is not simply the cost of fabricating the silicon itself, but includes all the steps required to obtain a tested and sellable device.

2.1.5 The computer market

The computer *market* is currently split in 5 different areas:

- *Personal Mobile Devices (PMDs)*: this area includes smartphones and tablet computers. Emphasis is on energy efficiency and real-time applications. Here, system price is around \$100 - \$1,000, with the microprocessor price at \$10 - \$100. Architectures are usually highly parallel (and in fact smartphones were the earliest devices to make heavy use of multi-core processors);
- Desktop computing: this area covers from PCs to workstations (and in some sense, even laptops). The main target of this area is to optimize the price-performance ratio. The system price is around \$300 - \$2,500, and of course the microprocessor price follows \$50 - \$500. Power is much less important than before, as we don't use batteries but power off the grid;
- Servers: these systems provide larger-scale and more reliable computing services. The main parameters of products in this area are:
 - Availability;
 - Scalability;
 - Throughput.

System price is around \$5,000 - \$10,000,000, with microprocessor price around \$200 - \$2,000;

- *Clusters / Warehouse Scale Computers (WSC)*: systems that offer "Software as a Service (SaaS)" or "Platform as a Service (PaaS)". The emphasis is on availability, price-performance, and power consumption. *Supercomputers* make up a part of this category, where the emphasis is on floating-point performance and fast internal networks. The system price is high at around \$100,000 - \$200,000,000, with the microprocessor price around \$50 - \$250 (because at times we might use off the shelf processors in massively parallel setups);
- Embedded computers: this area is a major application area of the computer market. It covers all special-purpose computer-based applications (from microwaves to coffee machines, from automotive to videogames). Adopted microprocessors vary from cheap low-end 8-bit processors to very efficient (and expensive) high-end processors. The system price ranges from \$10 - \$100,000, with microprocessor price at around \$0.01 - \$100 (possibly extremely cheap).

Special requirements often existing in this area are:

- Real-time performance requirements;
- Memory minimization;
- Power consumption minimization;
- Reliability constraints.

Embedded problems are often solved resorting to one of the following "standard" solutions:

- Standard processor, custom logic, custom software. We call the combination of standard processors with custom logic **ASIC** (*Application Specific Integrated Circuit*);
- Standard processor, custom software;
- Standard DSP, custom software.

Programmable devices (such as **FPGAs**) are playing a growing role.

2.1.6 Power

Continuous increase in system complexity and device integration often leads to problems with **power consumption**. Power consumption may be critical under two aspects:

- **Power**: *static* power, consumed when a device is just running without doing work, and *dynamic* power, which mainly depends on the frequency devices run at (and also the square power). Until now it has been dominated by dynamic power, i.e., that consumed by each transistor when switching between different states.
- The **dynamic** power for each transistor can be evaluated by the usual formula:

$$P_{dynamic} = \frac{1}{2} f V^2 C_{eq}$$

- Whereas **static** power can be evaluated by:

$$P_{static} = VI$$

and amounts around 50% of power consumption.

For this very reason, voltage continuously dropped in the last years.

- **Energy** consumed: it is given by:

$$E = C_{eq} V^2$$

It is mainly of interest for *portable* devices, where we are draining a battery (these include mobile phones and laptops, like the one the author of this notes is writing on, which will likely shut down if a power outlet is not found stat).

2.1.7 Dependability

Dependability is the quality of the system to deliver a correct service. Dependability of computer systems is traditionally very high, but it can be lowered by:

- Bugs in the design of the hardware;
- Bugs in the software;
- Defects in the hardware (introduced by the manufacturing process);

- Faults happening during the product operation.

Many **safety critical** systems exist. In the past, possible misbehaviors of computer systems were critical in some areas, such as

- Space;
- Avionics;
- Nuclear plants control.

In more recent years, computer-based systems expanded to other safety-critical areas, such as:

- Rail-road traffic control
- Automotive
- Biomedical
- Telecommunications.

Metrics for dependability are defined as:

- *Mean Time To Failure (MTTF)* or *Failures In Time (FIT)*, which is the reciprocal. 1 FIT equals 1 failure in one billion hours;
- *Mean Time To Repair (MTTR)*;
- *Mean Time Between Failures (MTBF)*.

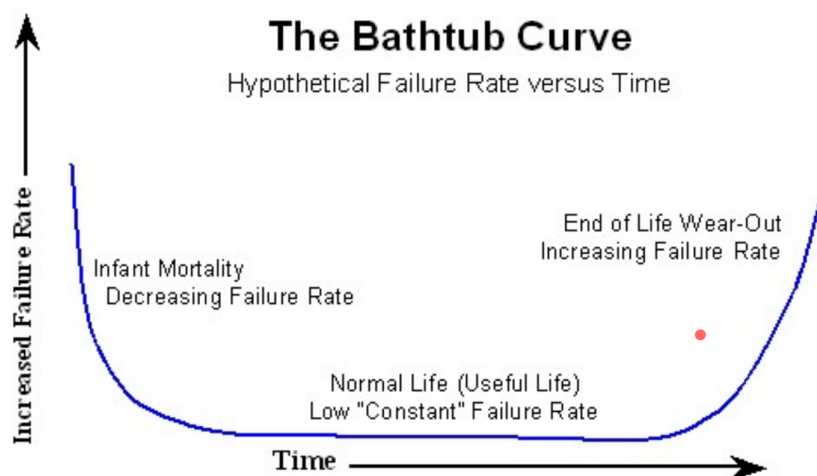
The three measures are related by the formula:

$$MTBF = MTTF + MTTR$$

Finally, availability is the probability that a system works correctly in a generic time instant (usually measured in FITs).

The **bathub curve** describes the probability of faults given the age of a component. We give it as a final example of how to take dependability into account.

- The *infant mortality* zone is where faults of components show themselves early. Controlled *burn-in* can partially solve this problem;
- The *normal life* zone is where expect to spend time using devices;
- The *end of life* zone is where failures hidden in components which otherwise worked fine start appearing. This is of particular interest to data centers where components are heavily stressed.



2.1.8 Reproducibility

Going back to the topic of *performance*, we introduce **reproducibility**. Information about execution times on benchmarks should allow reproducibility. This means reporting detailed information about:

- Hardware (system configuration);
- Software (OS, compiler, program);
- Program input.

A common issue is comparing and summarizing performance:

- **Problem 1:** I know the performance of one machine on a set of programs: which is its global performance?
- **Problem 2:** I know the performance of two machines on the same set of programs: which is their relative performance?

A number of metrics have been proposed, such as total execution time, arithmetic mean and weighted means.

2.1.9 Ahmdal's law

Definition 2.2: Ahmdal's law

We define the *speedup* from a given architecture enhancement as:

$$\text{speedup} = \frac{\text{performance with enhancement}}{\text{performance without enhancement}}$$

The speedup resulting from an enhancement depends on two factors:

- $\text{fraction}_{\text{enhanced}}$: the fraction of the computation time that takes advantage of the enhancement;
- $\text{speedup}_{\text{enhanced}}$: the size of the enhancement on the parts it affects.

We can compute the execution time after the speedup as:

$$\text{execution time}_{\text{new}} = \text{execution time}_{\text{old}} \cdot ()$$

Applying the definition of speedup, we get:

$$\text{speedup} = \frac{\text{performance with enhancement}}{\text{performance without enhancement}} = \frac{1}{(1 - \text{fraction}_{\text{enhanced}}) + \frac{\text{fraction}_{\text{enhanced}}}{\text{speedup}_{\text{enhanced}}}}$$

Ahmdal's law is particularly useful in the study of parallel architectures, as it expresses the speedup given by optimizing parallelizable parts of computing tasks. More specifically, we use it to explain why we can never have $S = N$ with N processor cores.

We can say that there will always be intervals of sequential execution (T_{seq}), spent executing instructions that cannot be parallelized, such as I/O operations, conditional statements, inherently sequential algorithms, etc... Then the speedup S can be defined as:

$$S = \frac{T_1}{T_{\text{seq}} + \frac{T_1 - T_{\text{seq}}}{N}}$$

with T_1 being the full execution time, i.e., the speed-up gain comes only from the parallelizable parts of the program ($T_1 - T_{\text{seq}}$), while the sequential parts (T_{seq}) cannot benefit significantly from parallelism.

The problem is that, taking the limit for $N \rightarrow \infty$, we obtain:

$$\lim_{N \rightarrow \infty} S = \lim_{N \rightarrow \infty} \frac{T_1}{T_{\text{seq}} + \frac{T_1 - T_{\text{seq}}}{N}} = \frac{T_1}{T_{\text{seq}}}$$

Thus, as N increases, the speed-up becomes dominated by the sequential parts: the parallelizable portions are progressively reduced, while the sequential portions remain and ultimately become the limiting factor for the overall system performance.

2.1.10 The CPU performance equation

Let's talk about measuring the time required to execute a program. There are several possible approaches:

- By observing the real system;
- By simulation;
- By applying the **CPU performance equation**.

Definition 2.3: CPU performance equation

CPU execution time is given by:

$$\text{CPU Time} = \left(\sum_i \text{CPI}_i \times \text{IC}_i \right) \times T_C$$

where:

- CPI_i is the number of clock cycles required by instruction i ;
- IC_i is the number of times instruction i is executed in the program;
- T_C is the clock period.

This equation is quite basic, as it doesn't take any processor optimizations into account. Still, it is a feasible metric for processor performance estimation, before the processor is even actually implemented.

2.2 Instruction Set Principles

Every processor owns an *Instruction Set Architecture (ISA)*. The ISA provides the information about the processor required by programmers and compilers to produce correctly working code:

- The set of available *instructions*;
- Instruction *encodings*;
- Available *registers*;
- etc...

Different processors may share the same ISA (e.g., Intel and AMD processors, or processors of the same family).

The Instruction Set Architecture (ISA) is basically how the computer is seen by the programmer or the compiler. There are many alternatives possible for the ISA designer,

any choice modifies the way to encode the instructions. The different alternatives may be evaluated in terms of:

- Processor performance;
- Processor complexity;
- Compiler complexity;
- Code size;
- Power consumption;
- etc...

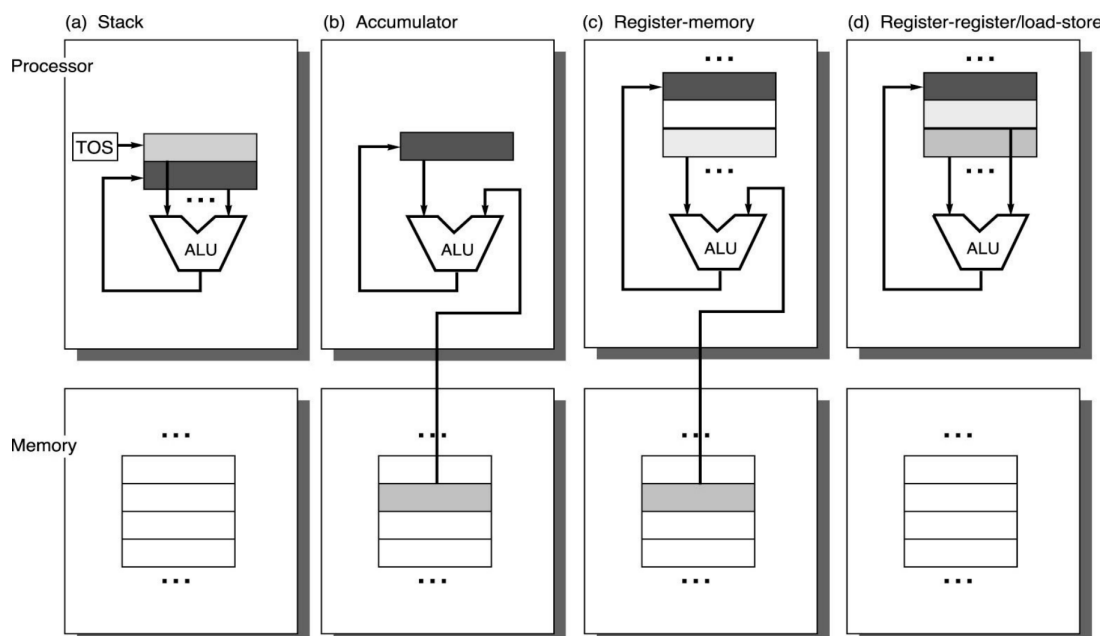
Different product areas may assign different weights to the above parameters.

An ISA specification is particularly important because it is usually the only set of information released by a company about a processor. Of course, the internal implementation details are trade secrets, but the ISA specification is made available in order to allow programmers to develop for a certain platform.

2.2.1 CPU taxonomies

CPUs (and their ISAs) are often classified according to the type of their internal storage:

- Stack;
- Accumulator;
- Registers:
 - Register-memory, as in x86;
 - Register-register (load-store). Basically all RISC architectures are of this kind;
 - Memory-memory (no real cases).



Currently, all processors are *General-Purpose Register (GPR)* machines. The reason for this is:

- Registers are faster than memory;
- Registers are easier for a compiler to use (or better, to *optimize*).

CPUs can also be classified according to:

- Typical number of operands per ALU instruction (2 or 3). In RISC-V and ARM, this is almost always 2;
- Typical number of memory operands per ALU instruction (from 1 to 3, for architectures that feature memory operands in the first place in instructions rather than load/store).

2.2.2 Memory addressing

We have some orthogonal characteristics relating to memory *addressing*:

- **Little Endian** vs. **Big Endian**: processors usually address memory in bytes, organized into words. The order of the bytes within a word matters:
 - *Little Endian* puts the byte with the lowest address (X...X000) at the least significant position. The address of the data is that of the least significant byte;
 - *Big Endian* puts the byte with the lowest address (X...X000) at the most significant position. The address of the data is that of the most significant byte.
- **Aligned** vs. **misaligned** accesses:
 - *Aligned* means allowing only aligned accesses not straddling word (usually 2 byte) bounds. It is a limitation;
 - *Disaligned* access, on the other hand, requires: hardware overhead and performance overhead.

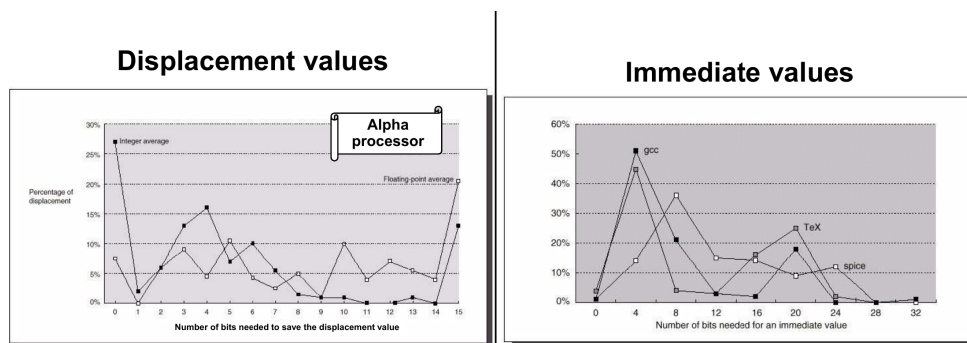
We shall also note that in GPR machines an addressing mode specifies a constant, a register, or a memory location (through its effective address).

We remember the addressing modes noted in section 1.2.5 for x86. Some of these are shared with ARM and RISC-V RISC architectures (mainly register mode, with destination register, immediate and displacement modes for load/store operations).

By carefully choosing the addressing modes, one can obtain some important consequences:

- Reducing the number of instructions;
- Increasing the CPU architecture complexity;
- Increasing the average *Cycles Per Instruction (CPI)*.

An interesting thing to note is the size devoted to immediate operands in immediate instructions. That is, how large can the immediate value be, when it is embedded in the addressing part of the instruction? Note that memory displacements basically are a special case of immediate operands, and as such fall in the same category and have the same problem. In fact, we had already noted this issue in section 1.2.5.



An experimental evaluation can be performed. From the plots, compiled from a DEC alpha processor for displacement and for typical compiled applications for immediates, we get that:

- Displacement, immediate and register indirect modes represent from 75% to 99% of the addressing modes;
- The address size for displacement mode should be from 12 to 16 bits (75% to 99% of the displacements);
- The size of the immediate field should be at least 8 or 16 bits (50% and 80% of the cases, respectively).

RISC-V uses the solution of doing 12 bits immediates. As we previously stated, bigger immediates have to be transferred to register in multiple instructions. This tradeoff is made valid by the simplicity gains of the processor architecture.

2.3 Operations in the instruction set

Clearly, a key component of an ISA is the **instruction set** made available to the programmer. Different categories of instructions exist:

Operator type	Examples
Arithmetic and logical	Integer arithmetic and logical operations: add, and, subtract, or
Data transfer	Loads-stores (move instructions on machines with memory addressing)
Control	Branch, jump, procedure call and return, traps
System	Operating system call, virtual memory management instructions
Floating point	Floating-point operations: add, multiply
Decimal	Decimal add, decimal multiply, decimal-to-character conversions
String	String move, string compare, string search
Graphics	Pixel operations, compression/decompression operations

We should be making the common case fast: not all the instructions are executed with the same frequency! When designing and implementing an instruction set, the most commonly executed instructions should be made faster than others.

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Let's continue talking about instruction set architectures. We recall that this part is a preliminary step in the design of every processor. The instruction set itself is, we said, the minimum set of information that the manufacturing company provides to users (programmers) of its processors. All the rest of the design remains confidential to the processor company (logical and physical architecture, ...).

Control flow instructions determine the control flow in the program. They can be distinguished in four categories:

- **Conditional branches:** this involves a comparison of values, which if evaluated to a true value, triggers a jump to some other location in code. Otherwise nothing happens;

- **jumps**: these are jumps to arbitrary code which always happen;
- Procedure **calls**: these involve saving some state about the CPU before jumping;
- Procedure **returns**: these involve recovering the state saved by procedure calls.

Conditional branches are by far the most frequently executed control flow instructions.

Control flow instructions affect the value of the *program counter* register, which points to the current instruction. The destination address is normally specified as a displacement with respect to the current value of the program counter. In this way:

- We save bits, since the target instruction is often close to the source one;
- The code is position-independent (**PIC**, *Position Independent Code*).

We should make a note about **register** based indirect jumps and procedure calls. They allow:

- To write code including jumps whose target is not known at compile time;
- To implement case or switch statements
- To support dynamically shared libraries (i.e., libraries which are loaded only when called)
- to support virtual functions (i.e., calling different functions depending on the data type).

We not show a table which compares jumps in multiple architectures.

Name	Examples	Test	Advantages	Disadvantages
Condition code (CC)	80x86, ARM, PowerPC, SPARC, SuperH	Tests special bits set by ALU operations, possibly under program control	Sometimes condition is set for free.	CC is extra state. Condition codes constrain the ordering of instructions because they pass information from one instruction to a branch
Condition register/limited comparison	Alpha, MIPS	Tests arbitrary register with the result of a simple comparison (equality or zero tests)	Simple	Limited compare may affect critical path or require extra comparison for general condition
Compare and branch	PA-RISC, VAX, RISC-V	Compare is part of the branch. Fairly general compares are allowed (greater than, less than)	One instruction rather than two for a branch	May set critical path for branch instructions

Let's now focus on **procedures**. When calling a procedure, as we said, some information need to be saved:

- The **return address**: this is the address of the instruction *right after* the call instruction. *Where* this address is stored depends on the architecture:

- x86 performs *stack* saving, accessing memory at the address pointed to by the stack pointer and decrementing it. This is referred to as *pushing* the address: *popping* the address involves incrementing the stack pointer and recovering the address;
- Architectures like RISC-V implement **JAL** (*Jump And Link*), where the return address is stored in a register. This saves one memory access for leaf procedure calls (that is procedures which don't call other procedures). If more calls are needed stack or other register saving can be performed manually.
- The accessed registers, based on whether our **ABI** (*Application Binary Interface*) defines *caller saving* or *callee saving* for certain registers (both can be defined, on different registers).
 - **Caller saved** registers are meant to be saved before the procedure call, if the caller intends to keep their value;
 - **Callee saved** registers are meant to remain unchanged after a procedure call. The called procedure can implement this by saving registers itself (through other registers or stack pushing), or by not using them at all.

To summarize instruction sets, we can say that:

- Few instructions are responsible for most of the execution time: load, store, add, subtract, move, register-register, and, shift, compare equal, compare not equal, branch, jump, call, and return.
- PC-relative branch displacements of at least 8 bits are the best choice.
- Register-indirect and PC-relative addressing can also be used in procedure call and return

3.0.1 Instruction set encodings

Instructions have to be encoded in a format the processor can *fetch* and *decoded* before executing. Instruction set **encoding** depends on:

- Which instructions compose the *instruction set*;
- Which *addressing modes* are supported (we have talked about register and memory addressing modes in section 2.2.2).

When a high number of addressing modes is supported, an address specifier field is used to specify the addressing mode and the registers which are possibly involved (this is the CISC philosophy). When the number of addressing modes is low, they can be encoded together with the opcode (this is the RISC philosophy).

We have 3 main types of instruction encoding:

- **Variable** encoding, where instructions have no fixed size and can grow based on immediates, displacements, ... This is the approach used by x86 and VAX architectures. Other characteristics are:
 - Supports any number of operands;
 - Instructions have a variable length;
 - Lower performance;
 - Minimum code size.
- **Fixed** encoding, where instructions are fixed in size and have fields which usually mean the same thing across instructions. This is the approach used by many RISC architectures, such as RISC-V, ARM, MIPS, PowerPC and SPARC.

Other characteristics are:

- Fixed number of operands;
 - Address specifier included in the opcode;
 - Fixed instruction length;
 - Maximum performance;
 - Larger code size.
- **Hybrid** encoding, where instructions are mostly fixed, but shorter instructions (often called *compressed* instructions) might appear for memory saving. This solves the problem of many RISC architectures, where the additional instructions that have to be included to perform operations grow code size (we talked about this in 1.2.4). This is the approach used by many compressed extensions for RISC architectures, such as RISC-V compressed (RV32IC), ARM Thumb2, microMIPS, and even older architectures like IBM 360/370.

Other characteristics are:

- Multiple formats (specified by the opcode);
- Allow trading-off between code size and performance.

We show a summarizing figure:

Operation and no. of operands	Address specifier 1	Address field 1	...	Address specifier n	Address field n
----------------------------------	------------------------	--------------------	-----	--------------------------	----------------------

(A) Variable (e.g., Intel 80x86, VAX)

Operation	Address field 1	Address field 2	Address field 3
-----------	--------------------	--------------------	--------------------

(B) Fixed (e.g., RISC-V, ARM, MIPS, PowerPC, SPARC)

Operation	Address specifier	Address field	
Operation	Address specifier 1	Address specifier 2	Address field
Operation	Address specifier	Address field 1	Address field 2

(C) Hybrid (e.g., RISC-V Compressed (RV32IC), IBM 360/370, microMIPS, Arm Thumb2)

We note that, when designing encodings, the designer should address several conflicting issues:

- The code size;
- The size of the Instruction Set, the number of addressing modes, and the number of registers;
- The complexity of the fetch and decoding hardware.

We should also note that nowadays, assembly-level programs are now produced by compilers, only. This eases the pressure off programmers, but means that the CPU desi-

gner and the compiler writer must interact and cooperate.

For instance, choosing which variables have to be put in which registers and when, is one of the crucial optimization phases in a compiler. This problem is based on graph coloring and can be solved easily if the number of registers is high (> 16). Of course, maximum performance would be achieved with an infinite number of registers, but this cannot be achieved as silicon is limited.

Optimizing variable access time by allocating variables to registers is only possible for those stored in the stack or for global variables in memory. It is impossible for variables belonging to the heap, due to the aliasing problem (i.e., the access to a variable is done through pointers).

To summarize, the CPU architect can help the compiler developer by:

- Making the frequent case fast and the rare case correct;
- Regularity;
- Providing primitives, not solutions;
- Simplifying trade-offs among alternatives;
- Providing instructions that bind the quantities known at compile time as constants.

The baseline recommendations are those of:

- **Orthogonality**, that is designing instructions that do not care about the *registers* which are used, or other specific constraints. Many famous instruction sets such as x86 are *not* orthogonal: for instance, multiplication and division instructions only operate on a specific subset of registers, in specific roles;
- **Simplicity**,
- Having at least 16 registers;

3.1 RISC-V introduction

We now give an introduction to the **RISC-V**. This and *ARM* will be the two architectures we cover during the course.

RISC-V is an open standard **ISA** (*Instruction Set Architecture*), having a few instructions in 3 different versions: 32, 64 and 128 bits. The fact that the ISA is fully open means that it is:

- Extremely easy and cheap to implement;
- Customizable and extensible.

The RISC-V ISA itself is defined as a base ISA (RV32I) plus extensions and user customizations. The base extensions feature things such as multiplication, protection, floating point operations, ...

Today, RISC-V is:

- Widely thought in universities;
- Increasingly adopted by companies (e.g., in the automotive domain);
- Strongly supported by the European Commission.

3.1.1 Technical characteristics

Let's get technical. Some generalities about RISC-V are:

- Freely licensed open standard;
- **Load/store** architecture;

- Fixed **32-bit** instruction set (whether RV32I, RV64I or RV128I is implemented);
- Additional **ISA extensions**, for example for security, floating point, and vector operations;
- The supported **addressing modes** are:
 - Displacement;
 - Immediate;
 - Register indirect.

The supported **data types** are:

- Byte (8 bits);
- Half Words (16 bits);
- Words (32 bits);
- Double Words (64 bits);
- 32-bit single precision floating-point;
- 64-bit double precision floating-point.

In regards to **encoding**, RISC-V allows a **variable** encoding in multiples of 16-bit blocks if code size reduction or specific implementations are needed.

Fixed-length 32-bit instructions are used in the base versions:

- 32-bit instructions have their lowest two bits set to 11;
- 16-bit instructions may have their lowest two bits equal to 00, 01, or 10.

In the case of 32-bit instructions, code memory alignment is a constraint; in other cases, the alignment is relaxed.

A little-endian memory system is the default choice.

3.1.2 Extensions

It counts with three base instructions sets plus a variety of optional extensions. For example, a 64-bit RISC-V possible ISA is defined as RV64IMAFD, known as RV64G, general-purpose scalar instruction set. The 32-bit version is called RV32G.

A table of common extensions is shown below.

Name of base or extension	Functionality
RV32I	Base 32-bit integer instruction set with 32 registers
RV32E	Base 32-bit instruction set but with only 16 registers; intended for very low-end embedded applications
RV64I	Base 64-bit instruction set; all registers are 64-bits, and instructions to move 64-bit from/to the registers (LD and SD) are added
M	Adds integer multiply and divide instructions
A	Adds atomic instructions needed for concurrent processing; see Chapter 5
F	Adds single precision (32-bit) IEEE floating point, includes 32 32-bit floating point registers, instructions to load and store those registers and operate on them
D	Extends floating point to double precision, 64-bit, making the registers 64-bits, adding instructions to load, store, and operate on the registers
Q	Further extends floating point to add support for quad precision, adding 128-bit operations
L	Adds support for 64- and 128-bit decimal floating point for the IEEE standard
C	Defines a compressed version of the instruction set intended for small-memory-sized embedded applications. Defines 16-bit versions of common RV32I instructions
V	A future extension to support vector operations (see Chapter 4)
B	A future extension to support operations on bit fields
T	A future extension to support transactional memory
P	An extension to support packed SIMD instructions: see Chapter 4
RV128I	A future base instruction set providing a 128-bit address space

3.1.3 Registers

The RV64G register bank consists of a register bank (or register file) of 32 elements, that is 32 integer registers $x_0, x_1, \dots, x_{31}$. The same amount of registers is available for floating point single (32-bit) and/or double precision (64-bit).

Register x_0 (called *register zero*) always returns the value 0, and cannot be written to. One reason for having such a register is shown below (3.1.4).

Other registers have roles defined by the ABI (though when doing bare metal we choose to respect them or not). We show a figure below of these roles.

Register	ABI Name	Description	Register	ABI Name	Description
x0	zero	hardwired zero	x16	a6	function argument 6
x1	ra	return address	x17	a7	function argument 7
x2	sp	stack pointer	x18	s2	saved register 2
x3	gp	global pointer	x19	s3	saved register 3
x4	tp	thread pointer	x20	s4	saved register 4
x5	t0	temporary register 0	x21	s5	saved register 5
x6	t1	temporary register 1	x22	s6	saved register 6
x7	t2	temporary register 2	x23	s7	saved register 7
x8	s0 / fp	saved register 0 / frame pointer	x24	s8	saved register 8
x9	s1	saved register 1	x25	s9	saved register 9
x10	a0	function argument 0 / return value 0	x26	s10	saved register 10
x11	a1	function argument 1 / return value 1	x27	s11	saved register 11
x12	a2	function argument 2	x28	t3	temporary register 3
x13	a3	function argument 3	x29	t4	temporary register 4
x14	a4	function argument 4	x30	t5	temporary register 5
x15	a5	function argument 5	x31	t6	temporary register 6

3.1.4 Addressing modes

Addressing modes for data transfers are *immediate* and *displacement*, both counting with an encoded field of 12 bits.

- **Immediate** addressing is done through the usual syntax, remembering that we have 3 registers per instruction (destination, source 1 and source 2, as per RISC ABIs): `addi x1, x2, 32;`
- **Displacement** is specified on indirect register addressing in load/store instructions: `lw x1, 32(x2)`. Let us also note that the `w` in `lw` stands for "word", as in load/store instructions in x86 fashion need their operand size specified in the opcode mnemonic. Of course a displacement of 0 equals to register indirect addressing. An interesting thing is how absolute displacements are specified: we have to include a register for all load instructions, and we can use just the 12 bit displacement as an address if we make that address `x0` (which we know always evaluates to 0).

3.1.5 Instruction formats

Instruction formats are an integral part of the RISC-V ABI. The *green card* available at https://www.cs.sfu.ca/~ashriram/Courses/CS295/assets/notebooks/RISCV/RISCV_CARD.pdf remains an extremely useful resource.

31	27	26	25	24	20	19	15	14	12	11	7	6	0	
funct7				rs2		rs1		funct3		rd		opcode		R-type
imm[11:0]						rs1		funct3		rd		opcode		I-type
imm[11:5]				rs2		rs1		funct3		imm[4:0]		opcode		S-type
imm[12 10:5]				rs2		rs1		funct3		imm[4:1 11]		opcode		B-type
imm[31:12]										rd		opcode		U-type
imm[20 10:1 11 19:12]										rd		opcode		J-type

Each instruction belongs to one of 5 formats, which specifies the semantics of fields within the instruction encoding. We see that fields like the instruction *opcode*, and the *funct3* and *funct7* values are shared between most formats.

- **R-type**: these are *register-to-register* instructions.

Field	Decription
opcode	7 bit primary operation opcode
rd	5 bit specifier for destination register
funct3	Specifies the type of operation (add / and / ...)
rs1	5 bit specifier for first register
rs2	5 bit specifier for second register
funct7	Selects a specific suboperation from a main operation (for instance sign extending and non-sign extending shifts)

- **I-type:** these are instructions with *immediates*.

Field	Decription
opcode	7 bit primary operation opcode
rd	5 bit specifier for destination register
funct3	Specifies the type of operation (addi / andi / ...)
rs1	5 bit specifier for first register
immediate	12 bit signed immediate used for logical operands, arithmetic signed operands, load/store address byte offsets, and PC-relative branch signed instruction displacement. Note that this is <i>sign extended</i> , as all other immediates in RISC-V! (for simplicity).

- **S-type:** these are *store* instructions.

Field	Decription
opcode	7 bit primary operation opcode
rs1	5 bit specifier for register to add to the 12 bits immediate
rs2	5 bit specifier for register to copy in memory
funct3	Specifies the size of the operation
imm fields	These two values compose the 12-bit signed value.

- **B-type:** these are *branch* (basically conditional additions affecting the program counter) instructions.

Field	Decription
opcode	7 bit primary operation opcode
rs1	5 bit specifier for first register to compare
rs2	5 bit specifier for second register to compare
funct3	Specifies the type of comparison
imm fields	These two values compose the 12 bit signed value.

- **U-type:** these are so-called *upper immediate* instructions, used to target the upper 20 bits 12 bit immediate instructions can't reach.

Field	Decription
opcode	7 bit primary operation opcode
rd	5 bit specifier for destination register (usually the program counter)
imm fields	These two values compose the 20 bit signed value.

- **J-type:** these are *jump* instructions.

Field	Description
opcode	7 bit primary operation opcode
rd	5 bit specifier for destination register (usually the program counter)
imm fields	These two values compose the 20 bit signed value.

3.1.6 Available instructions

The instructions available on the RV32G, grouped by function:

- Load and store
- ALU operations
- Branches and Jumps
- Floating Point
- Miscellaneous

In the following some of the most used instructions and pseudo-instructions are described.

Load/store: as we know, RISC-V processors use a load-store architecture. The main memory is accessed only through load and store instructions.

Example instruction	Instruction name	Meaning
ld x1,80(x2)	Load doubleword	$\text{Regs}[x1] \leftarrow \text{Mem}[80 + \text{Regs}[x2]]$
lw x1,60(x2)	Load word	$\text{Regs}[x1] \leftarrow_{64} \text{Mem}[60 + \text{Regs}[x2]]_0^{32} \text{ ##}$ $\text{Mem}[60 + \text{Regs}[x2]]$
lwu x1,60(x2)	Load word unsigned	$\text{Regs}[x1] \leftarrow_{64} 0^{32} \text{ ##} \text{Mem}[60 + \text{Regs}[x2]]$
lb x1,40(x3)	Load byte	$\text{Regs}[x1] \leftarrow_{64} (\text{Mem}[40 + \text{Regs}[x3]]_0^{56} \text{ ##}$ $\text{Mem}[40 + \text{Regs}[x3]])$
lbu x1,40(x3)	Load byte unsigned	$\text{Regs}[x1] \leftarrow_{64} 0^{56} \text{ ##} \text{Mem}[40 + \text{Regs}[x3]]$
lh x1,40(x3)	Load half word	$\text{Regs}[x1] \leftarrow_{64} (\text{Mem}[40 + \text{Regs}[x3]]_0^{48} \text{ ##}$ $\text{Mem}[40 + \text{Regs}[x3]])$
flw f0,50(x3)	Load FP single	$\text{Regs}[f0] \leftarrow_{64} \text{Mem}[50 + \text{Regs}[x3]] \text{ ##} 0^{32}$
fld f0,50(x2)	Load FP double	$\text{Regs}[f0] \leftarrow_{64} \text{Mem}[50 + \text{Regs}[x2]]$
sd x2,400(x3)	Store double	$\text{Mem}[400 + \text{Regs}[x3]] \leftarrow_{64} \text{Regs}[x2]$
sw x3,500(x4)	Store word	$\text{Mem}[500 + \text{Regs}[x4]] \leftarrow_{32} \text{Regs}[x3]_{32..63}$
fsw f0,40(x3)	Store FP single	$\text{Mem}[40 + \text{Regs}[x3]] \leftarrow_{32} \text{Regs}[f0]_{0..31}$
fsd f0,40(x3)	Store FP double	$\text{Mem}[40 + \text{Regs}[x3]] \leftarrow_{64} \text{Regs}[f0]$
sh x3,502(x2)	Store half	$\text{Mem}[502 + \text{Regs}[x2]] \leftarrow_{16} \text{Regs}[x3]_{48..63}$
sb x2,41(x3)	Store byte	$\text{Mem}[41 + \text{Regs}[x3]] \leftarrow_8 \text{Regs}[x2]_{56..63}$

Pseudo-instructions are not actual RISC-V instructions but they are often more convenient for the programmer. The assembler converts them to real RISC-V instructions (one or more).

One important pseudo-instruction is `li`, which stands for *load immediate*. As we know this operation is not straightforward in RISC-V because immediates are only 12 bits.

- For small immediates (which fit into 12 bits) the conversion is straightforward:

```

1 li t0, 4
2
3 # translates to:
4
5 addi t0, zero, 4

```

- For big immediates (which don't fit into 12 bits) the conversion needs an upper immediate instruction:

```

1 li t0, 0xFEDC8765
2
3 # translates to:
4
5 lui t0, 0xFEDC8      # upper 20 bits
6 addi t0, t0, 0x765   # lower 12 bits

```

ALU operations are performed on operands held in the processor registers (we also called them register-to-register operations).

Instruction types are:

- Immediate and three-operand instructions;
- Shift instructions;
- Multiply and divide instructions if the M extension is available.

Inst	Name	FMT	Opcode	funct3	funct7	Description (C)	Note
add	ADD	R	0110011	0x0	0x00	rd = rs1 + rs2	
sub	SUB	R	0110011	0x0	0x20	rd = rs1 - rs2	
xor	XOR	R	0110011	0x4	0x00	rd = rs1 ^ rs2	
or	OR	R	0110011	0x6	0x00	rd = rs1 rs2	
and	AND	R	0110011	0x7	0x00	rd = rs1 & rs2	
sll	Shift Left Logical	R	0110011	0x1	0x00	rd = rs1 << rs2	
srl	Shift Right Logical	R	0110011	0x5	0x00	rd = rs1 >> rs2	
sra	Shift Right Arith*	R	0110011	0x5	0x20	rd = rs1 >> rs2	msb-extends
slt	Set Less Than	R	0110011	0x2	0x00	rd = (rs1 < rs2)?1:0	
sltu	Set Less Than (U)	R	0110011	0x3	0x00	rd = (rs1 < rs2)?1:0	zero-extends

Some interesting pseudo-instructions we note are **seqz** and **snez**:

- seqz** sets a destination register if the source register is equal to 0:

```

1 seqz x1, x2
2
3 # translates to:
4
5 sltiu x1, x2, 1 # if (x2 == 0) x1 <- 1
6                # else x1 <- 0

```

- snez** sets a destination register if the source register is not equal to 0:

```

1 snez x1, x2
2
3 # translates to:
4
5 sltu x1, x0, x2 # if (x2 != 0) x1 <- 1
6                # else x1 <- 0

```

Branches and jumps: these comprise control transfer instructions (which are mainly unconditional), including:

- PC-relative jumps;
- Jump to register;
- Conditional branches.

jal	Jump And Link	J	1101111			rd = PC+4; PC += imm	
jalr	Jump And Link Reg	I	1100111	0x0		rd = PC+4; PC = rs1 + imm	

- j label:** a PC relative jump having a $\pm 1\text{MiB}$ range. It is a pseudo-instruction:

```

1 j label
2
3 # translates to:
4
5 jal x0, label

```

- **jal** label: a PC relative jump and link having a $\pm 1\text{MiB}$ range. It is a pseudo-instruction which assumes x1 to be the return address:

```

1 jal label
2
3 # translates to:
4
5 jal x1, label

```

- **jal** x3, label: a PC relative jump and link having a $\pm 1\text{MiB}$ range;
- **jr** x3: an indirect jump to register with $\pm 2\text{KiB}$ plus register range. It is a pseudo-instruction:

```

1 jr x3
2
3 # translates to:
4
5 jalr x0, 0(x3)

```

- **jalr** x3: an indirect jump to register with $\pm 2\text{KiB}$ plus register range. It is a pseudo-instruction which assumes x1 to be the return address:

```

1 jalr x3
2
3 # translates to:
4
5 jalr x1, 0(x3)

```

- **jalr** x3, x2, 0: an indirect jump and link to register. Saves the return value in the destination register.
- **call** sub1: calls a named subroutine. It is a pseudo-instruction which assumes x1 to be the return address:

```

1 call sub1
2
3 # translates to:
4
5 auipc x1, sub1 # top 20 bits of subroutine address
6 jalr x1, offset (x1)

```

- **ret**: returns from a subroutine. It is a pseudo-instruction which assumes x1 to be the return address:

```

1 ret
2
3 # translates to:
4
5 jalr x0, 0(x1)

```

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4.1 Pipelining

By studying RISC-V, we figured out that the CPU reads a sequence of fixed size instructions from memory, executing them one by one. We now introduce an implementation technique whereby multiple instructions are overlapped in execution: we call this technique **pipelining**. In a pipeline, different units (called pipe *stages* or *segments*) are (at any given time) completing different parts of different instructions in parallel.

Different stages of the pipeline perform different fixed functions on instructions. For instance, we might have (in an undefined implementation) that the execution of each instruction may be composed of at most five clock cycles:

- Instruction fetch cycle (IF)
- Instruction decode/register fetch cycle (ID)
- Execution/effective address cycle (EX)
- Memory access/branch completion cycle (MEM)
- Write-back cycle (WB).

4.1.1 Pipeline metrics

Let's define some *metrics*.

The **throughput** of a pipelined processor is the number of instructions which exit the pipeline in the time unit.

All the pipeline stages are synchronized (they proceed to executing a new task all together): the time for executing one step is called **machine cycle**, and normally corresponds to one *clock cycle*. The length of the machine cycle is determined by the slowest stage of the pipeline.

Lastly, we define **CPI** as *Clock Cycles per Instruction*.

In an ideal pipeline, all the stages are perfectly balanced (i.e., they require the same time). The throughput of an ideal pipeline (i.e., the number of instructions completed in a given time period) is:

$$\text{throughput}_{\text{pipelined}} = \text{throughput}_{\text{unpipelined}} * n$$

where n is the number of stages.

4.1.2 Pipeline performance

Pipelining increases the processor *throughput* without making single instructions execution faster. In fact, single instruction processing is made slower due to the pipeline control *overhead*. The depth of a pipeline is limited by:

- The need for balanced stages;
- PPipelining overhead (pipeline register delay and clock skew).

4.2 Pipeline hazards

Hazards are situations that prevent an instruction from executing during its designated clock cycle. There are three classes of hazards:

- **Structural hazards**: coming from resource conflict;

- **Data hazards:** an instruction depends on the result of a previous instruction;
- **Control hazards:** depend on pipelining branches and other instructions that change the PC.

Let's see ways to deal with these hazards.

4.2.1 Stalling

One way of dealing with hazards is to force the pipeline to **stall**, i.e., to block instructions for one or more clock cycles.

When an instruction is stalled:

- The instructions following the stalled instruction are also stalled;
- The instructions preceding the stalled instruction continue.

This can be easily achieved by modifying the buffer registers, making them able to *conserve* the previous value (this will be the stage which caused the hazard), and introducing an empty instruction to propagate onwards.

A stall causes the introduction of a "*bubble*" in the pipeline, which has to propagate through the whole pipeline.

This is the simplest way to solve pipeline hazards, as waiting it out (eventually) solves all problems.

4.2.2 Structural hazards

Structural hazards may happen when some pipeline unit is not able to execute all the operations scheduled for a given cycle. This is usually tied to limited resources in the processor.

Some common examples are:

- A given unit is not able to complete its task in one clock cycle;
- The pipeline owns only one register-file write port, but there are cycles in which two register writes are required;
- The pipeline refers to a single-port memory, and there are cycles in which different instructions would like to access to the memory together.

Solution

Removing structural hazards requires adding new hardware or improving the existing one. The designer has to trade-off between performance (add more resources or keep stalling) and cost (add more resources or keep the ones you have), basing on the frequency of occurrence of structural hazards.

This calls for optimization. Let's consider *load structural hazards*: these happen when two instructions contemporarily try to make a memory access to a single-port memory. The clear solution is to split instruction and data memory.

Assume that:

- 40% of instructions access the memory;
- The machine with the structural hazard has a clock 1.05 times faster than the one without.

We want to measure the performance gain.

- For the machine without structural hazard:

$$\text{Average Instruction Time}' = CPI \times T_C$$

where T_C is the clock period.

- For the machine with structural hazard:

$$\text{Average Instruction Time}'' = CPI_{\text{ideal}} + \text{Inst. freq} \times \text{Stall penalty}$$

computed using $T'_C = T_C \times 1.05$.

4.2.3 Data hazards

Overlapping the execution of instructions, as it is done by pipelining, changes the order of read/write accesses to operands: This can result in:

- Wrong results;
- Undeterministic behavior.

We call these **data hazards**.

Let's consider the code fragment:

```

1 add r1, r2, r3 # r1 write
2
3 sub r4, r1, r5 # r1 read from here onwards !
4 and r6, r1, r7
5 or  r8, r1, r9
6 xor r10, r1, r11

```

The problem is that the ADD instruction writes the result in R1 in the WB stage.

- The SUB instruction reads R1 in its ID stage, two clock cycles before the one in which R1 is made available (one to feed R! to WB and one to let WB finish): this produces a wrong result;
- The AND instruction reads R1 in its ID stage, one clock cycle before the one in which R1 is made available: this produces a wrong result;
- The OR instruction reads R1 in its ID stage, in the same clock cycle in which R1 is made available: this produces a correct result, provided that the write accesses to the register file are made before the read accesses. From now onwards, until we write R1 again, the results are correct (XOR is correct).

Another scenario in which we encounter data hazards is because of *interrupt effects*. If an interrupt occurs during the execution of a critical piece of code (from the point of view of data hazards), correctness may actually be restored (as instructions leave the pipeline as the interrupt is handled). This may cause an undeterministic behavior.

Solution

The typical solution to data hazards is that of **forwarding**. A special hardware in the datapath detects when a previous ALU operation should write the register corresponding to the source of the current ALU operation. In this case, the hardware selects the ALU result as the ALU input rather than the value from the register file.

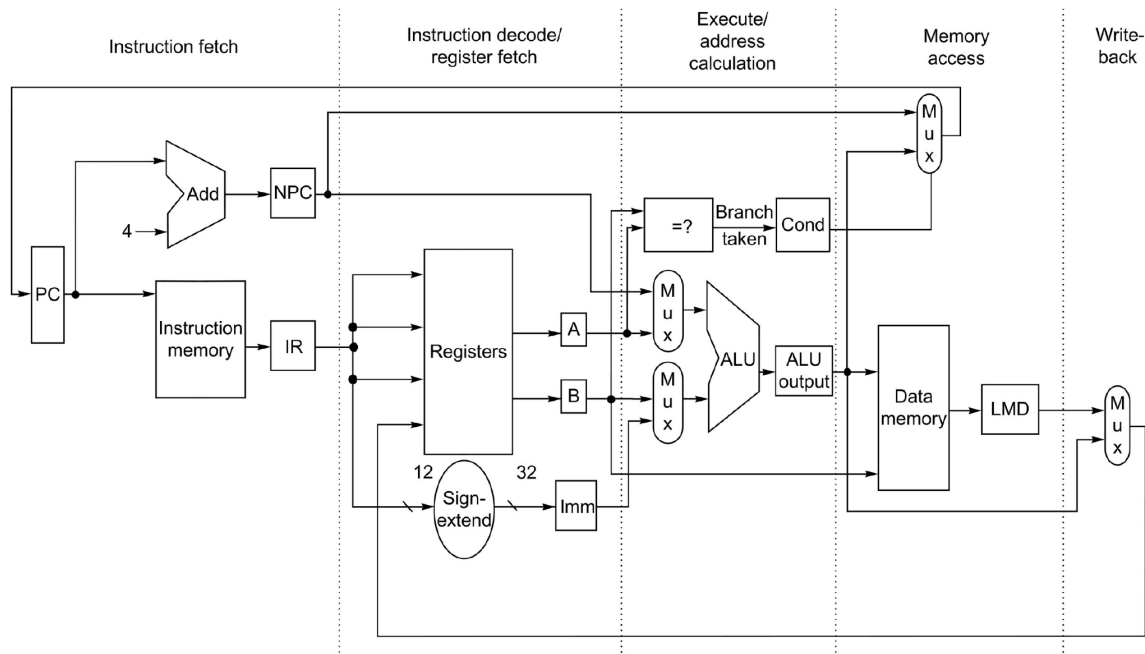
For this to be implemented, the hardware must be able:

- To forward a data from any of the previously started instructions (provided that they didn't already write the data in its final location);
- Not to forward anything, if the following instruction is stalled, or an interrupt occurred.

Forwarding can also be **generalized**: in order to always avoid stalling, it should be made possible between any pipeline register to any input of any functional unit.

4.3 Datapath model

Let's look at a simplified **datapath** schematic to explain pipelines with more detail.



4.3.1 Instruction fetch

The **instruction fetch** stage has the job of fetching the next instruction from program memory and feeding it to the next stage.

As we can see, we use the **PC** (*Program Counter*) to index instruction memory and put the instruction data into the **IR** (*Instruction Register*). Then we increment the program counter by 4 bytes (the size of a RISC-V instruction) and put that into an **NPC** (*New Program Counter*) register.

4.3.2 Instruction decode

We then have to **decode** the instruction.

A number of fields are extracted from the instruction encoding (we saw that RISC-V is highly regular). This allows for *fixed-field decoding*: decoding can be performed while registers are read.

- Parts of the instruction that index *registers* index the **register file**. In this example we index two registers, *A* and *B*;
- The immediate operand is also extracted from the instruction encoding, sign extended and sent to the next stage.

We can say that this stage performs both **decoding** and **register fetching**.

4.3.3 Instruction execution

At this point we can **execute** the instructions.

This is performed through the **ALU** (*Arithmetic Logic Unit*), which is basically a big multiplexer of different computation circuits.

We also have to distinguish between input registers and immediates (through multiplexers). Let's look at a couple of instruction types as an example:

- *Memory reference*: $ALU\ output \leftarrow A + Imm$. Note that we store the target *address* of the memory access! For loads we will use this directly, for stores we will use B and write the ALU output;
- *Register-Register ALU instruction*: $ALU\ output \leftarrow AfuncB$
- *Register-Immediate ALU instruction*: $ALU\ output \leftarrow AfuncImmediate$
- *Branch*: $ALU\ output \leftarrow NPC + Immediate << 2$, $Cond \leftarrow (A == B)$

Another thing to note is if the instruction is a branch one: if it is, we have to evaluate a condition and use it to multiplex between NPC and a calculated value (here the value of register A).

We can say that this stage performs both **execution** and **address calculation**.

4.3.4 Memory access

An extra stage is given by the **memory access** stage.

- If a (**load**) instruction requires it, we use the ALU output to address memory and load the needed contents in the **LMD** (*Loaded Memory Data*) register;
- If a (**store**) instruction requires it, we use the B register to address memory, and write the ALU output to it.

This is also the stage where we perform the *branch completion cycle*, that is decide if we should update the NPC or keep it at $PC + 4$.

More clearly:

- *Memory reference*: $LMD \leftarrow Mem[ALUOutput]$ (loads) or $Mem[ALUOutput] \leftarrow B$ (stores);
- *Branch*:

$$cond ? PC \leftarrow ALU\ output, \text{ otherwise } PC \leftarrow NPC$$

4.3.5 Write-back cycle

The last step to perform is the **write-back**.

Based on the instruction type, we update the register file with the obtained result:

- *Register-to-register* or *register-to-memory* instructions feed the ALU output to the register file;
- *Load* instructions feed the LMD to the register file.

More clearly:

- *Register-Register ALU instruction*: $Regs[IR11.7] \leftarrow ALU\ output$;
- *Register-Immediate ALU instruction*: $Regs[IR11.7] \leftarrow ALU\ output$;
- *Load instruction*: $Regs[IR11.7] \leftarrow LMD$;

In this discussion we skipped a lot of details about the implementation, for instance how the multiplexers are driven, etc... We assume these details to be (more or less) easy to synthesize as combinatorial or sequential networks once an actual architecture is defined.

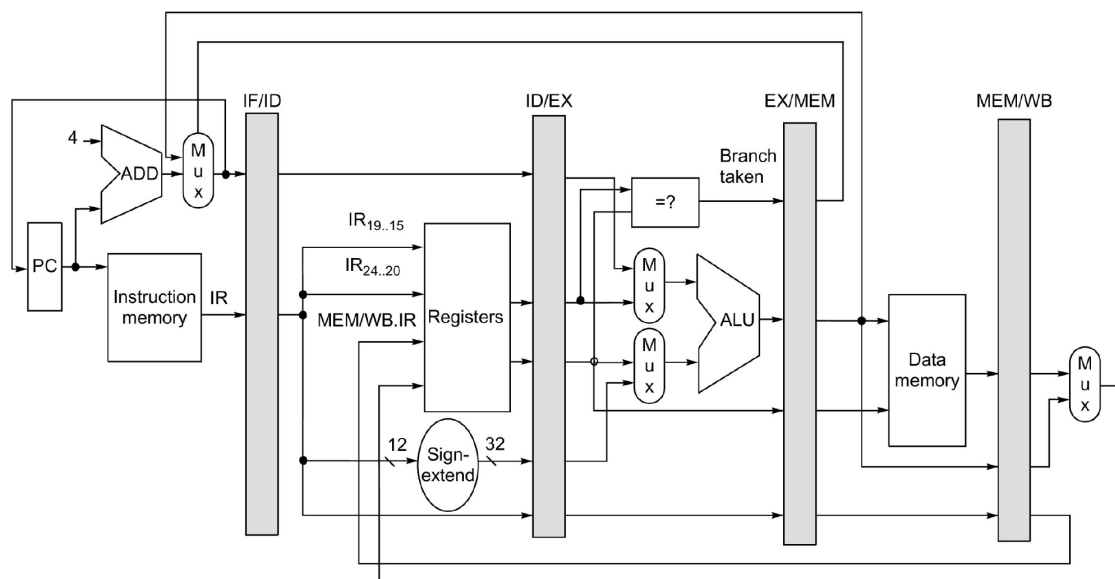
In the described architecture, all instructions require 5 clock cycles, except branch instructions, which require 4 clock cycles. Optimizations could be done for reducing the

average CPI: as an example, the ALU instructions could be completed during the memory access cycle. Also, hardware resources could be optimized by avoiding duplications (e.g., for ALUs, and memory). An alternative single-clock architecture (i.e., executing an instruction per clock cycle) can also be considered, but in that case a simple control unit is required to produce the signals required by the datapath.

4.3.6 Pipelined version

A different solution can be using a **pipelined** version, where a new instruction is started at each clock cycle. Different resources work on different instructions at the same time, and at every clock cycle, each resource can be used for one purpose, only. This means that:

- Separate instruction and data memories (i.e., caches) must be used;
- The register file is used in two stages: for reading (second half of the cycle) in ID and for writing (first half of the cycle) in WB. It must be designed to satisfy these requests during the same clock cycle.
- The PC must be changed in the IF stage, and branch handling must be considered;
- *Pipeline registers* must be added between stages to act as buffers.



4.3.7 Structural hazard control

We don't go in depth on **structural hazard** control as we saw that it depends on the resources available to the processor.

4.3.8 Data hazard load control

Let's first go over the **causes** of data hazards.

Causes

A hazard is created whenever there is dependence between instructions, and they are close enough that the overlap caused by pipelining would change the order of access to an operand. In general, this can happen for:

- **Register operands;**
- **Memory operands:** this is possible if:
 - Accesses to memory by load and store are not made in the same stage;
 - Execution can proceed while an instruction waits for a cache miss to be solved.

A problem is that not all data hazards can be solved through data forwarding. Focusing on **loads**, let's consider the code fragment:

```

1 ld  r1, 0(r2)  # r1 loaded
2
3 sub r4, r1, r5 # r1 used
4 and r6, r1, r7
5 or  r8, r1, r9

```

Here, when `sub` enters the execution stage, the data memory access is just being initiated on instruction `ld`. The forwarding path here goes back in time, therefore it is impossible to implement. This means it's impossible to forward: our only solution is a stall.

Load control

Implementing the **control** for these bubbles requires that at each clock cycle:

- All tests for detecting a possible data hazard concerning an instruction are performed when this is in the ID stage
- If a data hazard is detected, two actions can be alternatively taken:
 - The appropriate forwarding is activated;
 - The instruction is stalled before entering the stage where operands are not available (i.e., before being issued).

The control circuitry used to handle loads is called **LID** (*Load Interlock Detection*).

Let's look at a table of possible hazard conditions:

Situation	Example fragment	Action
No dependence	LD R1,45(R2) DADD R5,R6,R7 DSUB R8,R6,R7 OR R9,R6,R7	No hazard possible because no dependence exists on R1 in the immediately following three instructions.
Dependence requiring stall	LD R1,45(R2) DADD R5,R1,R7 DSUB R8,R6,R7 OR R9,R6,R7	Comparators detect the use of R1 in the DADD and stall the DADD (and DSUB and OR) before the DADD begins EX.
Dependence overcome by forwarding	LD R1,45(R2) DADD R5,R6,R7 DSUB R8,R1,R7 OR R9,R6,R7	Comparators detect use of R1 in DSUB and forward result of load to ALU in time for DSUB to begin EX.
Dependence with accesses in order	LD R1,45(R2) DADD R5,R6,R7 DSUB R8,R6,R7 OR R9,R1,R7	No action required because the read of R1 by OR occurs in the second half of the ID phase, while the write of the loaded data occurred in the first half.

From the table, we deduce that the following checks have to be performed when a load instruction is in the EX stage and another instruction exploiting the loaded value is

in the ID stage.

ID Opcode	IF Opcode Match	
Load	Register-Register ALU	ID/EX.IR[rt] == IF/ID.IR[rs]
Load	Register-Register ALU	ID/ EX.IR[rt] == IF/ID.IR[rt]
Load	Load, store, ALU immediate, branch	ID/EX.IR[rt] == IF/ID.IR[rs]

If operands match, a data hazard is detected and as a result, the control unit must insert a pipeline stall and prevent the instructions in IF and ID stages from advancing.

At this point, given an instruction currently in the ID stage, introducing a stall in the EX stage can be done by:

- Forcing a nop instruction in the ID/EX pipeline register (in RISC-V the nop corresponds to an `addi x0, x0, 0`);
- Forcing the IF/ID pipeline register to maintain the current value (*conserve*);
- Freezing the PC (maintain unaltered IF status).

4.3.9 Data hazard forwarding control

Let's now shift our attention to **forwarding**. Forwarding can be implemented:

- From the ALU or data memory output;
- To ALU inputs, data memory inputs, or the zero detection unit (branch instructions).

The forwarding logic must compare:

- The destination fields of the IR contained in the EX/MEM and MEM/WB registers;
- The source fields of the IR contained in the IF/ID, ID/EX and EX/MEM registers.

We see one final table regarding the forwarding conditions and the *comparisons* (checks):

Pipeline register of source	Opcode of source	Pipeline register of destination	Opcode of destination	Forwarding destination	Comparison
EX/MEM	Register-register ALU ID/EX ALU immediate	ID/EX	Register-register ALU, ALU immediate, load, store, branch	Top ALU input	EX/MEM . IR [rd] == ID/EX . IR [rs1]
EX/MEM	Register-register ALU ID/EX ALU immediate	ID/EX	Register-register ALU	Bottom ALU input	EX/MEM . IR [rd] == ID/EX . IR [rs2]
MEM/WB	Register-register ALU, ALU immediate, Load	ID/EX	Register-register ALU, ALU immediate, load, store, branch	Top ALU input	MEM/WB . IR [rd] == ID/EX . IR [rs1]
MEM/WB	Register-register ALU, ALU immediate, Load	ID/EX	Register-register ALU	Bottom ALU input	MEM/WB . IR [rd] == ID/EX . IR [rs2]